

Famous Paintings: Queries

Queries

-- Q1 List of all Norwegian artists

```
SELECT full_name, nationality
FROM artist
WHERE nationality = 'Norwegian'
```

	full_name text	nationality text
1	Hans Dahl	Norwegian
2	Edvard Munch	Norwegian

-- Q2 List of all French Naturalist artists

```
SELECT full_name, nationality, style
FROM artist
WHERE nationality = 'French' AND style = 'Naturalist'
```

	full_name text	nationality text	style text
1	Julien Dupré	French	Naturalist
2	Charles-Francois Daubigny	French	Naturalist

-- Q3 Top 5 nationalities by number of artists

```
SELECT nationality, count(artist_id)
FROM artist
GROUP BY nationality
ORDER BY count(artist_id) DESC
Limit 5
```

	nationality text	count bigint
1	French	115
2	American	83
3	Dutch	68
4	English	44
5	Italian	26

-- Q4 Top 10 artists by number of works

```
SELECT a.full_name, count(w.name)
FROM artist a
JOIN work w
ON a.artist_id = w.artist_id
GROUP BY 1
ORDER BY 2 DESC
LIMIT 10
```

	full_name text	count bigint
1	Pierre-Auguste Renoir	469
2	Claude Monet	378
3	Vincent Van Gogh	308
4	Maurice Utrillo	253
5	Albert Marquet	233
6	Henri Lebasque	200
7	Camille Pissarro	188
8	Louis Valtat	184
9	Maximilien Luce	174
10	Henri Le Sidaner	159

-- Q5 Smallest canvas sizes by area

```
SELECT width, height, (width * height) AS area
FROM canvas_size
ORDER BY area
LIMIT 10
```

	width integer	height integer	area integer
1	16	18	288
2	20	16	320
3	16	20	320
4	15	22	330
5	16	23	368
6	24	16	384
7	16	24	384
8	20	20	400
9	22	19	418
10	24	18	432

-- Q6 Works by Vincent Van Gogh

```
SELECT a.full_name, w.name
FROM artist a
JOIN work w
ON a.artist_id = w.artist_id
WHERE a.full_name = 'Vincent Van Gogh'
LIMIT 10
```

	full_name text	name text
1	Vincent Van Gogh	A Group of Cottages
2	Vincent Van Gogh	A Lane Near Arles, Side of a Country Lane
3	Vincent Van Gogh	A Pair of Leather Clogs
4	Vincent Van Gogh	A Path Through a Ravine
5	Vincent Van Gogh	A Peasant Woman Digging in Front of Her Cott...
6	Vincent Van Gogh	A Woman Walking in a Garden
7	Vincent Van Gogh	Agostina Sagatori Sitting in the Cafe' du Tamb...
8	Vincent Van Gogh	Apples
9	Vincent Van Gogh	Avenue with Flowering Chestnut Trees
10	Vincent Van Gogh	Bank of the Oise at Auvers

-- Q7 Most common art subjects across works

```
SELECT s.subject, count(a.artist_id)
FROM artist a
JOIN work w
ON a.artist_id = w.artist_id
JOIN subject s
ON s.work_id = w.work_id
GROUP BY 1
ORDER BY 2 DESC
LIMIT 10
```

	subject text	count bigint
1	Portraits	1070
2	Abstract/Modern ...	575
3	Nude	525
4	Landscape Art	495
5	Rivers/Lakes	480
6	Flowers	457
7	Still-Life	395
8	Seascapes	326
9	Marine Art/Mariti...	268
10	Horses	265

-- Q8 Sunday opening hours of museums in France

```
SELECT m.name, h.open, h.close
FROM museum m
JOIN museum_hours h
ON m.museum_id = h.museum_id
WHERE h.day = 'Sunday' AND m.country = 'France'
```

	name text	open text	close text
1	Museum of Grenoble	10:00:AM	06:30:PM
2	Musée du Louvre	09:00:AM	06:00:PM
3	Musée d'Orsay	09:30:AM	06:00:PM
4	Museum of Fine Arts of Nancy	10:00:AM	06:00:PM
5	Musée Marmottan Monet	10:00:AM	06:00:PM
6	Army Museum	10:00:AM	06:00:PM

-- Q9 Distribution of museums by website domain type

```
SELECT
COUNT(museum_id) AS total_museums,
CASE
    WHEN url LIKE '%.com%' THEN 'Commercial (.com)'
    WHEN url LIKE '%.org%' THEN 'Organization (.org)'
    WHEN url LIKE '%.edu%' THEN 'Education (.edu)'
    WHEN url LIKE '%.gov%' THEN 'Government (.gov)'
    WHEN url LIKE '%.net%' THEN 'Network (.net)'
    WHEN url LIKE '%.museum%' THEN 'Museum (.museum)'
    WHEN url IS NULL THEN 'No URL'
    ELSE 'Other'
END AS url_category
FROM museum
GROUP BY url_category
ORDER BY total_museums DESC;
```

	total_museums bigint	url_category text
1	29	Organization (.org)
2	22	Other
3	3	Education (.edu)
4	2	Government (.gov)
5	1	Museum (.museum)

-- Q10 Distribution of works by percentage price decrease

```
SELECT
  COUNT(work_id) AS num_works,
  CASE
    WHEN price_change = 0 THEN 'No price change'
    WHEN price_change >= -0.10 THEN '0-10% decrease'
    WHEN price_change >= -0.20 THEN '10-20% decrease'
    WHEN price_change >= -0.30 THEN '20-30% decrease'
    WHEN price_change >= -0.40 THEN '30-40% decrease'
    WHEN price_change >= -0.50 THEN '40-50% decrease'
    WHEN price_change >= -0.75 THEN '50-75% decrease'
    ELSE '75%+ decrease'
  END AS price_change_category
FROM (
  SELECT
    work_id,
    (sale_price::numeric - regular_price::numeric)
    / NULLIF(regular_price::numeric, 0) AS price_change
  FROM product_size
) t
GROUP BY price_change_category
ORDER BY 2;
```

	num_works bigint	price_change_category text
1	5	10-20% decrease
2	132	20-30% decrease
3	10928	30-40% decrease
4	91684	40-50% decrease
5	46	50-75% decrease
6	12	75%+ decrease
7	7540	No price change